Fertility and Housing Tenure Choice

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Housing and Fertility

Housing Market | Real Estate 4 minute read

US birth rate hits record low, housing costs weigh on family planning

The 2008 financial crisis pushed both housing and fertility

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News Article

US Birth Rate Warning Issued Over House Prices

Want to Boost Birthrates? Build More Homes

Policymakers looking to reduce childcare costs should work to expand housing supply.

POLICY

Will cheap housing lead to more babies?

The connection between housing and fertility rates has a missing piece.

Renting a home 'leads to lower birth rates'

Housing, Fertility, and Tenure Choice

Children increase how much physical space a household needs. Access to that space depends on tenure and wealth:

- Rental housing can limit the size of the home a family can occupy.
- Ownership expands the housing menu but requires liquid wealth for a down payment.
- Older owners may retain family-sized homes after their children leave.

How much does access to family-sized housing affect fertility?
Which housing constraints matter, and for whom?

This Paper

Need a framework where fertility, tenure, housing size, saving, and population are jointly determined.

First: a simplified model of family space and tenure

- Show when reallocating housing toward the young improves welfare at fixed fertility.
- Give a household condition for better housing access to raise fertility.

Then: a quantitative lifecycle model

- Add sequential fertility, income risk, saving, housing adjustment, and bequests.
- Distinguish births, resident persons, household heads, and housing demand.

Finally: discipline the model and use it to study demographic and housing-market dynamics.

Simple Theory

Environment and Preferences

A two-period OLG economy: Y_t young and O_t old share a fixed housing stock $\bar{H}$.

- Entrants draw (y_i^y, b_i, y_i^o) from F: young income, liquid wealth, and old income.
- Young choose consumption, housing, completed fertility, saving, and lifetime tenure.
- Old choose consumption, housing, and an estate; future income is known.

Flow utilities are:

$$u_t^y(c, h, n) = \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta_t \log n,$$

$$u^o(c^2, h^2, e) = \log c^2 + \gamma \log h^2 + \omega_B \log e.$$

Each child needs χ goods and κ housing; $n > 0$ is continuous. All utility weights are positive. Households discount future utility by β.

Owning adds $\xi_i \sim \text{Logistic}(\bar{\xi}, \sigma_\xi)$, independent of endowments. The warm-glow estate e does not determine entrant wealth.

Young Households

Goods and bonds trade externally at bond price $q \in (0,1)$. Housing costs P_t; mortgages finance at most share $\phi_t \in (0,1)$; the property-tax rate is τ^p.

Taking prices and the property-tax rebate T_t as given:

$$W_t^R(i) = \max_{c,h,n,a'} u_t^y(c,h,n) + \beta V_{t+1}^R(a';i)$$

$$c + qa' + qr_t h = y_i^y + b_i + T_t, \qquad a' \geq 0, \quad h \leq h_R^{\max},$$

$$W_t^O(i) = \max_{c,h,n,a'} u_t^y(c,h,n) + \beta V_{t+1}^O(a',h;i)$$

$$c + qa' + (1 + q\tau^p)P_t h = y_i^y + b_i + T_t,$$

$$qa' + \phi_t P_t h \geq 0, \qquad h \leq h_O^{\max}.$$

Here a' is next-period net financial wealth, including mortgage repayment. Rent r_t is paid at period end; $h_R^{\max} < h_O^{\max}$.

The equal property-tax rebate T_t, current income, and wealth are available before purchase.

Old Households and Tenure

An old household has net financial wealth a and income y_i^o; an owner also holds title to H units. Taking prices and rebates as given:

$$V_t^R(a;i) = \max_{c^2,h^2,e} u^o(c^2,h^2,e)$$

$$c^2 + qe + qr_t h^2 = a + y_i^o + T_t, \qquad h^2 \leq h_R^{\max},$$

$$V_t^O(a,H;i) = \max_{c^2,h^2,e} u^o(c^2,h^2,e)$$

$$c^2 + qe + qr_t h^2 = a + P_t H + y_i^o + T_t,$$

$$h^2 \leq h_O^{\max}, \qquad e \geq P_{t+1} h^2.$$

Financial saving $a^e \geq 0$ gives $e = q^{-1}a^e$ for renters and $e = q^{-1}a^e + P_{t+1}h^2$ for owners. Owners may buy or sell within $h_O^{\max}$.

Young own if $W_t^O(i) + \xi_i \geq W_t^R(i)$; this determines the owner probability $\pi_t^O(i)$. Tenure remains fixed thereafter.

Competitive Equilibrium

Given policies, F, and (Y_0, O_0, G_0), an equilibrium is a sequence of prices (P_t, r_t), rebates T_t, household choices, cohort sizes, and old distributions satisfying:

1. Young and old optimize at equilibrium prices; tenure follows the value comparison.
2. Rental entry: $qr_t = (1 + q\tau^p)P_t - qP_{t+1} > 0$.
3. Housing clears and property-tax revenue is rebated equally:

$$Y_t \bar{h}_t^y + O_t \bar{h}_t^o = \bar{H}, \qquad (Y_t + O_t)T_t = q\tau^p P_t \bar{H}.$$

4. Cohorts follow $Y_{t+1} = \nu \bar{n}_t Y_t$ and $O_{t+1} = Y_t$.
5. G_{t+1} is generated by young choices of (a', h) and tenure, together with y_i^o.

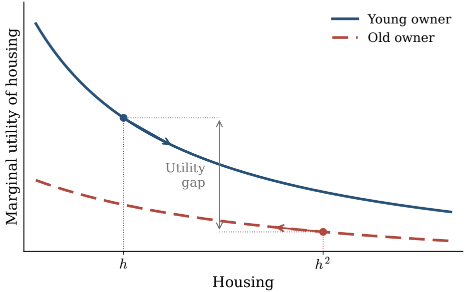
Bars are cross-sectional averages over endowments and tenure; ν converts children to entrants.

G_t is the old distribution of financial wealth, inherited housing, income, and tenure.

A positive steady state has constant prices, choices, distributions and cohort sizes, with $Y^* = O^*$ and replacement fertility $\bar{n}^* = 1/\nu$.

Housing Allocation and Welfare

At a stationary equilibrium, give each living household equal weight on its remaining utility.



With $s = h - \kappa n$, marginal housing utility is higher for a constrained young owner than for its stationary old counterpart:

$$\frac{\alpha}{s} > \frac{\gamma}{h^2}.$$

Utilitarian gain. Move a little housing from old to young, holding consumption, fertility, tenure, estates, and future allocations fixed. Old housing utility falls; total welfare rises.

Requires $\beta \geq q$, a positive share of owners with strictly binding mortgages, slack young and old housing caps, and positive old financial saving. The direct planner may relax finance. Conditions Proof

Housing and Fertility

At fixed tenure, prices, and other transfers, a current cash gift raises fertility. More space lowers the housing cost of children; consumption also matters.

For two chosen bundles at the same ϑ, if (c_1, h_1) can accommodate n_0:

$$n_1 \geq n_0 \quad \Longleftrightarrow \quad \frac{\chi}{c_1 - \chi n_0} + \frac{\alpha\kappa}{h_1 - \kappa n_0} \leq \frac{\chi}{c_0 - \chi n_0} + \frac{\alpha\kappa}{h_0 - \kappa n_0}.$$

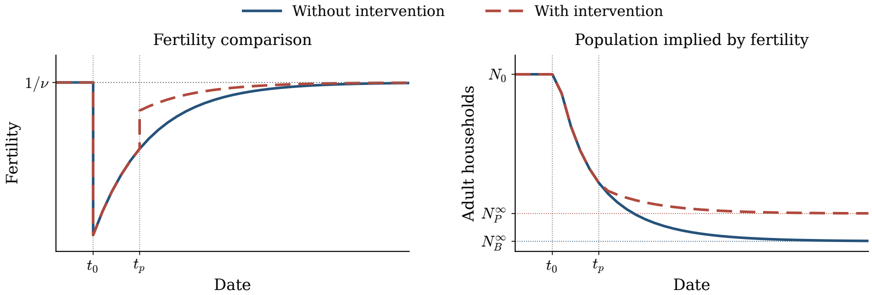
A rise in both consumption and housing is sufficient; otherwise their effects must be compared.

The current and old-age grants in the appendix raise treated owners' fertility at fixed prices and rebates. Aggregate effects also depend on tenure choices, housing-market clearing, and future cohort sizes.

Details

Fertility and Population Paths

An exogenous fall in ϑ_t at t_0 starts demographic adjustment. At t_p, a housing policy branches from the same inherited state as the baseline.



Higher fertility along the policy path makes later cohorts larger: $Y_{t+1} = \nu \bar{n}_t Y_t$.

Fertility paths are assumed. Both approach replacement and yield positive population limits by cohort accounting; equilibrium housing and fiscal conditions remain to be established.

Dated comparisons

Quantitative Model

Environment

- Households live for a finite number of periods in one housing market.
- Beginning-of-period state:

$$x = (b, z, a, h, n, m)$$

  - b: liquid net wealth; z: earnings state; a: age
  - h: inherited owner size, with $h = 0$ for renters
  - n: literal parity; m: number of dependent children at home
- Households choose whether to attempt the next birth, housing tenure and size, consumption, and saving.
- Earnings, conception, child maturation, and survival are stochastic.

The model keeps fertility and housing choices at the household level while tracking the persons and household heads they generate.

Preferences

Let m denote the number of dependent children currently at home.

Flow utility. Households value consumption, usable housing space, and children; $e(m)$ scales total household needs:

$$u_t(c, s; m) = e(m)^{\sigma-1} \frac{\left[c^{\alpha_0}\left(s - \bar{h}(m)\right)^{1-\alpha_0}\right]^{1-\sigma}}{1-\sigma} + \psi_t m.$$

Family-space needs and housing services are

$$\bar{h}(m) = h_1 \mathbf{1}\{m > 0\} + h_m m, \qquad s = \begin{cases} h^R, & \text{renter,} \\ \chi h, & \text{owner,} \end{cases} \qquad \chi > 1.$$

- h_1 is a discrete first-child space requirement; h_m is the additional requirement per child at home.
- Child needs disappear as children mature, while parity remains part of the household state.

Housing, Tenure, and Budget Constraints

Households choose housing inputs

$$h^R \in [0, \bar{h}^R], \qquad h' \in \{0\} \cup \mathcal{H}^O, \qquad h' h^R = 0.$$

Tenure is implicit: $h^R > 0$ means rent; $h' > 0$ means own.

Changing the inherited owner stock h to h' changes liquid resources by

$$\mathcal{T}_t(h, h') = \mathbf{1}\{h' \neq h\} \left[(1 - \psi^s) P_t h - P_t h'\right].$$

Renters ($h' = 0$)

$$c + r_t h^R + b' = R_b b + y(a, z) + \mathcal{T}_t(h, 0) + T_t.$$

Owners ($h^R = 0$)

$$c + (\delta_H + \tau_t^p) P_t h' + b' = R_b b + y(a, z) + \mathcal{T}_t(h, h') + T_t.$$

- A buyer needs cash of at least $(1 - \phi) P_t h'$, where ϕ is the financed share.
- The debt floor rules out new unsecured borrowing and tapers inherited debt with age.

Sequential Fertility and Child Aging

- During fertile ages, a household below terminal parity chooses whether to attempt the next birth.
- An attempt succeeds with age-specific probability π_a.
- A successful birth:

$$n' = n + 1, \qquad m' = m + 1,$$

  and the first successful birth pays the one-time cost F_1.
- Fertility taste shocks have scale κ_1 on the first-birth margin and κ_C on continuation margins.
- Each child at home matures independently; m falls as children leave, while parity n remains a stock.

A birth changes the household's space needs immediately, but its demographic effects propagate across cohorts.

Earnings and Bequests

Earnings

- Lifecycle earnings profile
- Persistent idiosyncratic risk
- Permanent-income heterogeneity
- Retirement income after working life
- Liquid return R_b

Warm-glow bequests

$$\mathcal{B}(w^e) = \theta_0 \frac{(\theta_1 + \max\{w^e, 0\})^{1-\sigma}}{1-\sigma}.$$

- Estate wealth includes liquid assets and net housing sale proceeds.
- Bequests affect saving and old-age housing retention.
- Entrant wealth is externally distributed: the model is not fully dynastic.

Housing Supply and Price Mapping

Occupied housing demand is the integral of physical rooms:

$$D_t = \int h_t(x)\, dG_t(x).$$

Housing supply is static and upward sloping:

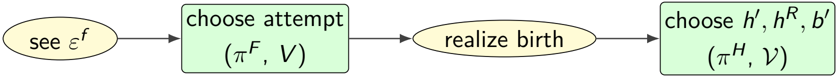
$$H_t^s = H_0 \left(\frac{P_t}{\bar{P}} \right)^{\eta}, \qquad D_t = H_t^s.$$

Housing is also an asset. Rent and the current purchase price satisfy

$$r_t + P_{t+1} = (R_b + \delta_H + \tau_t^p) P_t.$$

- P_t capitalizes the next-period resale value.
- r_t is the current renter price.
- Owner service utility χh does not create extra physical housing supply.
- The elasticity η determines how a demand shift is divided between prices and quantities.

The Household's Problem



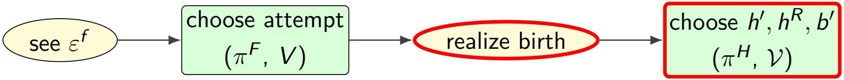
State.

$$x = (b, z, a, h, n, m), \qquad h \in \{0\} \cup \mathcal{H}^O.$$

- b: liquid wealth; z: earnings; a: age; h: inherited owner stock.
- n: parity; m: number of children currently at home.
- The model has one housing market, so location is suppressed.

Within the period: observe the fertility shock, choose whether to attempt, realize conception success, then choose tenure, rooms, consumption, and saving.

The Household's Problem: Housing and Saving

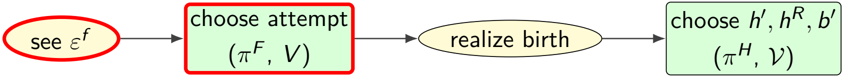


Conditional on the realized parity branch, the post-fertility value is

$$\mathcal{V}_t(x) = \mathbb{E}_{\varepsilon_t^H} \max_{h', h^R, b'} \left\{ u_t(c, s; m) + \varepsilon_t^H(h') + \beta s_a \mathbb{E}_t[V_{t+1}(x')] + \beta(1 - s_a)\mathcal{B}(w_{t+1}^e) \right\}.$$

- $h' h^R = 0$: a renter sets $h' = 0$; an owner sets $h^R = 0$.
- Current purchases occur at P_t and future sales at future prices.
- The budget, down-payment condition, and age-tapered debt floor restrict (h', h^R, b').
- Household policies use the date-$(t + 1)$ continuation value.

The Household's Problem: Fertility Choice



Let x^+ add one birth, and let π_a be conception success. Before the attempt taste shock:

$$W_t^0(x) = \mathcal{V}_t(x), \qquad W_t^A(x) = \pi_a\left[\mathcal{V}_t(x^+) - F_1\mathbf{1}\{n=0\}\right] + (1-\pi_a)\mathcal{V}_t(x).$$

The attempt probability is

$$\pi_t^F(x) = \frac{\exp\{W_t^A(x)/\kappa(n)\}}{\exp\{W_t^A(x)/\kappa(n)\} + \exp\{W_t^0(x)/\kappa(n)\}}, \qquad \kappa(n) = \begin{cases} \kappa_1, & n = 0, \\ \kappa_C, & n \geq 1. \end{cases}$$

Housing affects fertility through current child-space needs, tenure feasibility, and the continuation value of wealth and housing.

Persons, Heads, and Households

The demographic block distinguishes three distributions:

$$\underbrace{N_t}_{\text{resident persons}} \quad \Longrightarrow \quad \underbrace{H_t}_{\text{household heads}} \quad \Longrightarrow \quad \underbrace{G_t}_{\text{household economic states}} \quad .$$

- Births enter the person distribution immediately.
- Survival and migration move person cohorts across age and sex.
- Headship maps persons into the mass of household decision makers.
- The household transition moves wealth, earnings, tenure, parity, and child states within that head mass.

The total mass of the household distribution must equal the head mass implied by persons.

The Person-Cohort Law

Let $N_t^{g,a}$ denote resident persons of sex g and single year of age a.

The annual person law is

$$N_{t+1} = A_{t+1} N_t + f_{t+1} B_{t+1} + M_{t+1}.$$

- A_{t+1} ages existing cohorts using destination-age survival.
- $f_{t+1} B_{t+1}$ allocates newborn survivors by sex at age zero.
- M_{t+1} is the age–sex net-migration flow after survival.
- Model-period births are assigned to calendar cohorts before the annual law is iterated.

Births and household heads remain distinct objects throughout the transition.

The Household-Head Law

Fixed age–sex headship rates $\rho^{g,a}$ map persons into the target head stock:

$$H_{t+1}^{g,a} = \rho^{g,a} N_{t+1}^{g,a} = H_{t+1}^{\mathrm{surv},g,a} + F_{t+1}^{g,a} - D_{t+1}^{g,a} + H_{t+1}^{M,g,a}.$$

The household operator supplies a raw within-age distribution $\widetilde{G}_{t+1,a}$. Its mass is rescaled to the person-law head target:

$$G_{t+1,a}(x) = H_{t+1,a} \frac{\widetilde{G}_{t+1,a}(x)}{\int \widetilde{G}_{t+1,a}(x')\, dx'}.$$

- F and D are the minimum one-way formation or dissolution flow needed to attain $\rho^{g,a}$.
- Wealth, tenure, income, parity, and child-state composition are preserved within age.
- This is an accounting closure, not an estimated household-formation hazard.

Transition Equilibrium

A perfect-foresight transition is a path

$$\{V_t, \pi_t, G_t, N_t, H_t, P_t, r_t, T_t\}_{t=0}^{T}$$

such that, at every date:

1. households optimize given future prices and transfers;
2. births advance persons and heads, and the household distribution is rebalanced to head mass;
3. occupied housing equals static-elastic supply;
4. the asset-price/rent identity holds;
5. property-tax revenue equals the equal transfer paid to household heads.

Steady-State Equilibrium

A steady state is a constant tuple

$$(N^*, H^*, G^*, \pi^*, P^*, r^*, T^*)$$

that satisfies simultaneously:

- household optimality and household-distribution stationarity;
- the person-cohort fixed point under the stated survival and migration closure;
- head formation and exact agreement between head mass and household mass;
- housing-market clearing, asset pricing, the government budget, and demographic reproduction.

A finite path is not a steady state. Neither is a constant household distribution whose births fail to reproduce persons and heads.

What the Quantitative Model Adds

- Sequential first and continuation births rather than a one-time family-size choice.
- A lifecycle of earnings, saving, tenure adjustment, child maturation, retirement, and death.
- A discrete owner ladder and continuous rental choice with collateral and transaction costs.
- Bequests and late-life housing retention.
- Separate laws for resident persons, household heads, and household economic states.
- Endogenous housing prices, rents, fiscal transfers, and demographic feedback.

The quantitative model preserves the access mechanism while measuring its exposure and general-equilibrium consequences.

Quantification

Dated Calibration

The model is estimated at a point along the 2007–2023 transition.

1. Derive the 2007 fertility intercept so a constructed old benchmark has completed fertility 2.1.
2. Reweight only the age marginal to the 2007 ACS household-head distribution.
3. Estimate a linear fertility-preference change from 2007 through 2023.
4. Impose Census household totals and ACS head-age shares at five dates.
5. Match twelve moments, principally measured on the simulated 2023 state.

The household-total and age bridge is an input, not a fit. It is retired after 2023 and replaced by the person/head transition law.

Externally Calibrated Parameters

	Param	Value	Source or restriction
Timing	Model period	4 years	Maintained
	Entry, maximum age	18, 85	Maintained
Pref.	σ	2.0	Maintained
Finance	ϕ	0.80	20% buyer equity
	ψ^s	0.06	Housing literature benchmark
Housing	Owner menu	$\{2, 4, 6, 8, 10\}$	ACS rooms distribution
	Rental cap	6 rooms	ACS rental distribution
	η	1.75	Saiz benchmark
Entry	Wealth distribution	PSID	Childless renters, ages 18–24

Internally Calibrated Parameters

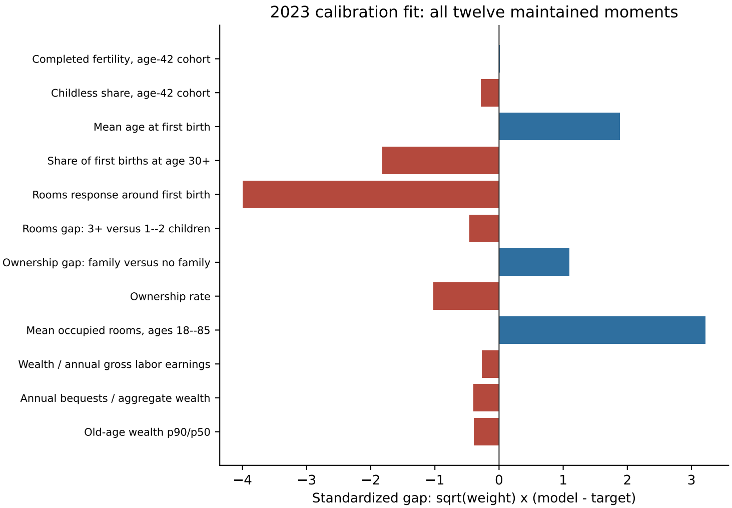
Param	Description	Estimate
β_{annual}	Annual discount factor	0.994959
κ_1	First-birth logit scale	2.267631
κ_C	Continuation-birth logit scale	1.765427
χ	Owner-service premium	1.033945
H_0	Housing-supply scale	16.381189
θ_0	Bequest scale	0.563219
θ_1	Bequest wealth shift	0.099707
h_m	Per-child room floor	0.315267
h_1	First-child room jump	0.199446
F_1	First-birth fixed cost	4.545167
$\Delta\psi_{2007:2023}$	Fertility-preference change	-0.350000

All eleven estimates are interior to their search bounds.

Calibration Targets

	Moment	Target	Model
Fertility	Completed fertility, women 40–44	1.918	1.918
	Childless share, women 40–44	0.188	0.186
	Mean age at first birth	26.045	26.327
	First births at age 30 or later	0.260	0.242
Housing	Four-year rooms contrast at first birth	0.720	0.380
	Rooms gap, 3+ versus 1–2 children	0.368	0.359
	Family ownership gap	0.168	0.177
	Ownership rate, ages 30–55	0.575	0.546
	Mean occupied rooms, ages 18–85	5.780	6.710
Wealth	Wealth / annual gross labor earnings	6.873	6.766
	Annual bequest flow / aggregate wealth	0.0088	0.0086
	Old wealth-to-income p90/p50	3.448	3.396

Calibration Fit



What the Calibration Does Not Match

The strict SMM loss is 36.099. Two housing moments account for 26.304, or 73% of the objective:

Moment	Target	Model
Four-year rooms contrast at first birth	0.720	0.380
Mean occupied rooms	5.780	6.710

- Fertility stocks, tenure, and wealth fit more closely than housing quantities.
- The weighted Jacobian has rank $10/11$ at the predeclared 10^{-3} threshold.
- Its condition number is $1{,}104$; the weakest direction loads primarily on the bequest shift θ_1.
- The estimate is reproducible and interior, but locally weakly identified and not a claim of global optimality.

From the 2023 Fit to the Transition

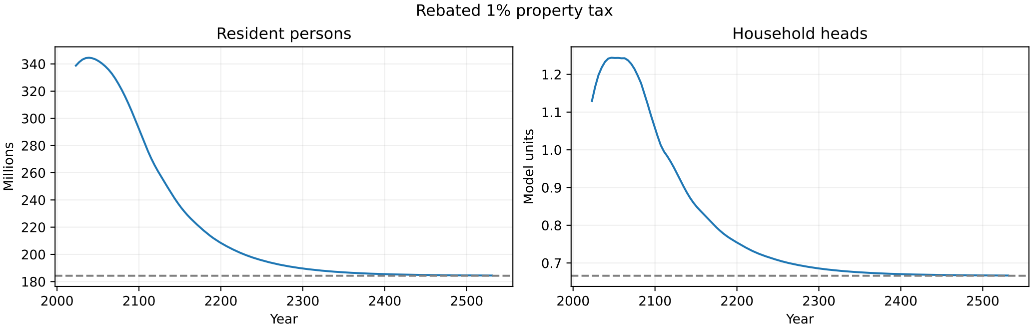
Object	Annual tax	Rebate	Role
Calibration status quo	1%	None	Fitted 2023 reference
Transition reference	1%	Equal per head	Accepted forward path
Policy counterfactual	2%	Equal per head	Same equilibrium closure

The accepted transition begins from:

338.7 million persons, 129.0 million heads, the fitted 2023 household state.

Supply is reanchored so the rebated 1% economy clears at the inherited quantity. The transition reference is rebated; it is not the unrebated calibration status quo.

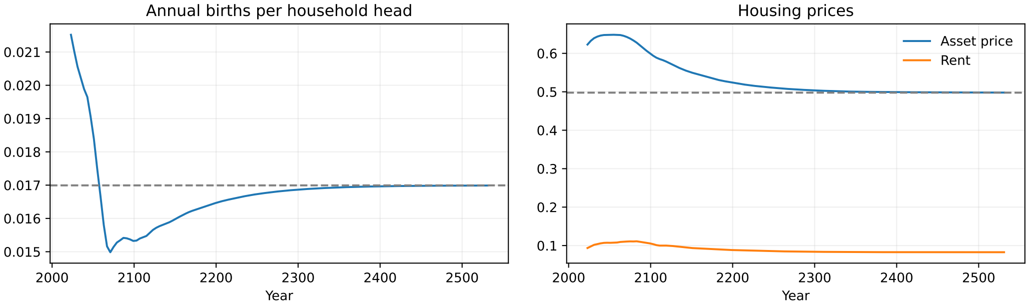
Demographic Momentum



Resident persons rise from 338.7 million in 2023 to 344.6 million in 2039. Household heads peak in 2047 at 142.1 million. Both stocks then contract toward the conditional terminal equilibrium.

An inherited age distribution generates decades of momentum even when subsequent births are low.

Prices and Births Move on Different Clocks



The asset price peaks in 2055. The renter price peaks in 2083 because it also reflects the next-period asset payoff. Annual births per head undershoot their terminal rate before converging back toward it.

Cohort composition, forward-looking capitalization, and household fertility need not turn at the same date.

The Conditional Terminal Equilibrium

Object	2023	Terminal	Change
Resident persons, millions	338.724	184.278	−45.6%
Household heads, millions	128.964	76.073	−41.0%
Asset price	0.6229	0.4977	−20.1%
Rent-equivalent price	0.0936	0.0825	−11.8%
Annual births per head	0.02152	0.01699	−21.0%

The terminal demographic renewal ratio is 0.5568. A constant positive population is sustained by the fixed age–sex migration flow.

This is a conditional fixed-migration equilibrium, not a forecast of the U.S. population level.

The Rebated Property Tax

The reform permanently raises the annual property-tax rate from 1% to 2% and returns all contemporaneous revenue equally across household heads:

$$T_t \int dG_t = \tau_{H,t} P_t Q_t^H, \qquad Q_t^H = \int h_t(x)\, dG_t(x).$$

- **Capitalization**: a higher carrying cost tends to reduce the asset price.
- **User cost**: the renter price depends on the tax and the next asset price.
- **Tenure and fertility**: down payments, owner products, and child-space choices adjust state by state.
- **Demographic scale**: changed births affect persons now and household heads only later.

All effects are general equilibrium: prices, tenure, births, population, and the per-head rebate evolve together.

Conclusion

- Children raise demand for rooms; rental size and down payments determine access to that space.
- Fertility, tenure, saving, and old-age housing retention must be solved jointly over the lifecycle.
- Births create persons; survival, migration, and head formation create household heads.
- The calibrated model fits fertility stocks, tenure, and wealth more closely than housing quantities.
- The accepted transition displays demographic momentum followed by slow convergence to a conditional fixed-migration equilibrium.

Housing policy can affect family formation today and housing demand generations later.

Thank You

Appendix

Young Household Problems and Finance

A prime denotes next-period net financial wealth. Let $p_t = qr_t$ denote the current cost of housing services. The two conditional problems are:

$$W_t^R(i) = \max_{c,h,n,a'} \; u_t^y(c,h,n) + \beta V_{t+1}^R(a';i),$$

$$c + qa' + p_t h = y_i^y + b_i + T_t, \quad a' \geq 0, \quad h \leq h_R^{\max},$$

$$W_t^O(i) = \max_{c,h,n,a'} \; u_t^y(c,h,n) + \beta V_{t+1}^O(a',h;i),$$

$$c + qa' + (1 + q\tau^p)P_t h = y_i^y + b_i + T_t,$$

$$qa' + \phi_t P_t h \geq 0, \quad h \leq h_O^{\max}.$$

If d is mortgage principal and k is current bond investment, then $0 \leq d \leq \phi_t P_t h$, $k \geq 0$, and $a' = (k - d)/q$. Eliminating k, d gives the owner's constraint.

Current income, wealth, and the rebate can finance purchase. All choices satisfy $c > \chi n$, $h > \kappa n$, and $n > 0$.

Old Household Problems and Financial Estates

Write $z_R = a + y_i^o + T_t$, $z_O = a + P_t H + y_i^o + T_t$, and $p_t = qr_t$. The old household solves:

$$V_t^R(a;i) = \max_{c^2,h^2,e} \; u^o(c^2,h^2,e), \quad c^2 + qe + p_t h^2 = z_R, \quad h^2 \leq h_R^{\max},$$

$$V_t^O(a,H;i) = \max_{c^2,h^2,e} \; u^o(c^2,h^2,e), \quad c^2 + qe + p_t h^2 = z_O,$$

$$h^2 \leq h_O^{\max}, \qquad e \geq P_{t+1}h^2.$$

The owner's cash budget is $c^2 + a^e + (1+q\tau^p)P_t h^2 = z_O$, with $a^e = q(e - P_{t+1}h^2) \geq 0$. At stationary prices, put $K = 1+\gamma+\omega_B$ and $p = (1-q+q\tau^p)P$. If both owner restrictions are slack:

$$c^2 = \frac{z}{K}, \qquad h^2 = \frac{\gamma z}{Kp}, \qquad e = \frac{\omega_B z}{Kq}, \qquad a^e = \frac{z}{K}\left(\omega_B - \frac{qP\gamma}{p}\right) > 0.$$

Positive financial saving requires $\omega_B(1-q+q\tau^p) > q\gamma$. Without a binding size cap, failure of this inequality makes the estate floor bind. The housing-welfare proof uses the positive-saving regime.

Stationary Existence and Price Bounds

Assume bounded endowments $w_{0i} = y_i^y + b_i \geq \underline{w} > 0$, $v_{0i} = y_i^o \geq 0$, and $0 \leq \tau^p < 2$. Define the cost coefficients and resource bounds:

$$d_p = 1 - q + q\tau^p, \quad d_L = 1 - \phi + q\tau^p, \quad d_{\max} = \max\{d_p, d_L\},$$

$$K = 1 + \gamma + \omega_B, \quad D = 1 + \alpha + \vartheta + \beta K,$$

$$\bar{M}_0 = \mathbb{E}_F(w_0 + qv_0), \quad \bar{T} = \frac{\tau^p \bar{M}_0}{(1-q)(2-\tau^p)}, \quad \bar{M} = \bar{M}_0 + (1+q)\bar{T}.$$

Sufficient fertility at low housing prices is ensured by:

$$h_R^{\max} > \frac{\kappa}{\nu}, \qquad \vartheta\nu > \frac{\chi D}{\underline{w}} + \frac{\alpha\kappa}{h_R^{\max} - \kappa/\nu}.$$

A positive stationary equilibrium then exists, and every such equilibrium obeys:

$$0 \leq T \leq \bar{T}, \qquad P_- \equiv \frac{\alpha\underline{w}}{D d_{\max} h_R^{\max}} < P < P_+ \equiv \frac{\nu\bar{M}}{d_p\kappa}.$$

Fertility is above replacement at the lower price and below it at the upper price. Continuity and the bounded rebate map give a stationary fixed point; these bounds do not require uniqueness.

Constrained Owners: Primitive Conditions

Use the preceding bounds, and let $\mathcal{S}$ be a positive-mass endowment group. Define:

$$\ell = \frac{d_L}{d_p}, \qquad C_{\min} = 1 + \alpha\ell + \vartheta \min\{1, \ell\}, \qquad k = \frac{D}{C_{\min}} - 1.$$

A sufficiently steep income profile makes its mortgage strictly restrictive:

$$qv_{0i} > kw_{0i} + \max\{k - q, 0\}\,\bar{T} \qquad (i \in \mathcal{S}).$$

Let $M_{\mathcal{S}} \geq \sup_{i \in \mathcal{S}}\{w_{0i} + qv_{0i} + (1+q)\,\bar{T}\}$. Require:

$$\omega_B d_p > q\gamma, \qquad h_O^{\max} > \frac{M_{\mathcal{S}}}{d_p P_-} \max\left\{1, \frac{\gamma}{qK}\right\}.$$

These inequalities make old financial saving positive and both owner housing caps slack.

Logistic tastes give $\mathcal{S}$ positive owner mass.

Without the mortgage constraint, desired cash expenditure is at least $(w + qv)C_{\min}/D > w$. The household cannot bring enough of its old resources forward. With $\beta \geq q$, these primitive restrictions imply the welfare result.

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Direct Utilitarian Reallocation

At stationarity, let $x = c - \chi n$, $s = h - \kappa n$, $p = d_p P$, $L = d_L P$, $w = y^y + b + T$, $v = y^o + T$, and $z = a' + Ph + v$. The young owner's restrictions are:

$$c + ph + qz = w + qv, \qquad c + Lh \leq w.$$

Let $\mu > 0$ be the cash-constraint multiplier and $m = K/z = 1/c^2$ the marginal utility of old resources. The household conditions imply:

$$\frac{\alpha}{s} - \frac{\gamma}{h^2} = \left(\frac{\beta}{q} - 1\right) pm + \mu L > 0.$$

A transfer of ε housing from the matched old owner to the young changes equal-weight welfare by:

$$\Delta \mathcal{W} = \alpha \log(1 + \varepsilon/s) + \gamma \log(1 - \varepsilon/h^2) > 0 \quad \text{for small } \varepsilon > 0.$$

Set $\Delta a' = -P\varepsilon$, $\Delta a^e = qP\varepsilon$, with cash $p\varepsilon$ from old to young. This preserves estates, young future resources, and aggregate bonds. Match owners by endowment across equal stationary cohorts. The planner relaxes private finance; the gain is local.

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Transfers with Committed Fertility and Tenure

Start from the constrained stationary owner group with $\beta \geq q$. An unexpected intervention occurs after fertility and tenure are committed, before housing purchase. Suppose:

$$\phi \geq q, \qquad \alpha(1+\omega_B) \leq \gamma \frac{1-\phi+q\tau^p}{1-q+q\tau^p}.$$

With $\lambda = \beta K/(qz)$, the following choices add ε adult space:

$$s_\varepsilon = s+\varepsilon, \qquad x_\varepsilon = \frac{L}{\alpha/(s+\varepsilon)-(p-L)\lambda}.$$

Implement them with current grant G and promised old-age grant J:

$$G = x_\varepsilon - x + L\varepsilon, \qquad qJ = (p-L)\varepsilon.$$

These lump-sum grants, additional to T, are nonnegative. Under the original mortgage rule, the young choose more housing and consumption; old resources z stay fixed.

The intervention requires fertility and tenure to be committed before housing choice.

Transfer Funding and Welfare

An uncapped old owner's housing demand is gz, where $g = \gamma/(Kp)$. Tax its matched counterpart ε/g to release exactly ε housing. Additional old owners strictly at their housing cap pay the residual:

$$R = G + qJ - \varepsilon/g = x_\varepsilon - x + p\varepsilon - \varepsilon/g \geq 0.$$

The capped funders have lower marginal utility of cash: $m_F < \gamma/(ph_O^{\max}) < m$. A primitive sufficient income bound is:

$$y_f^o > (\phi/q - 1)P_+ h_O^{\max} + \frac{Kd_p P_+ h_O^{\max}}{\gamma}.$$

With $A = \mathrm{d}x_\varepsilon/\mathrm{d}\varepsilon|_0$ and $\Lambda = 1/x$, the welfare gain is:

$$\mathcal{W}'(0) = (\Lambda - m)A + (\alpha/s - pm) + (m - m_F)(A + p - 1/g) > 0.$$

Positive funder mass and uniform strict margins are required. The government saves qJ to pay J. At the original price, housing clears and old estates finance higher consumption. Young old-age choices and subsequent cohorts are unchanged.

Fertility after a Grant

Private fertility satisfies the balance between its utility and goods-and-space costs:

$$\frac{\vartheta}{n} = \frac{\chi}{c - \chi n} + \frac{\alpha\kappa}{h - \kappa n}.$$

For the same grant pair (G, J), reoptimize all other choices at baseline fertility n_0. Adult consumption and space both rise. If $W_\varepsilon(n)$ is the profiled owner value:

$$W'_\varepsilon(n_0) = \chi\left(\frac{1}{x_0} - \frac{1}{x_\varepsilon}\right) + \alpha\kappa\left(\frac{1}{s_0} - \frac{1}{s_\varepsilon}\right) > 0.$$

Concavity therefore implies higher optimal fertility at fixed prices, rebates, and tenure. A current cash gift alone also raises conditional fertility across mortgage and old-estate regimes.

Across tenures, mean fertility for endowment type i changes by:

$$\Delta\bar{n}_i = \pi_1^O \Delta n_i^O + (1 - \pi_1^O)\Delta n_i^R + (\pi_1^O - \pi_0^O)(n_{i0}^O - n_{i0}^R).$$

Tenure composition can offset conditional gains. Once fertility is freed, the earlier tax-and-grant construction need not clear housing or preserve future cohorts.

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Dated Welfare and Population Comparisons

Let $m_{i,t+1}$ and $m_{j,t}$ be future and current old marginal utilities of cash. With housing caps and old estate floors slack, the dated housing gain requires:

$$p_t\left(\frac{\beta}{q}m_{i,t+1}-m_{j,t}\right)+\mu_{i,t}L_t>0,\qquad p_t=qr_t,\quad L_t=(1-\phi_t+q\tau^p)P_t.$$

The gap must be checked along the path. With identical initial cohorts:

$$\frac{Y_t^{\mathrm{policy}}}{Y_t^{\mathrm{baseline}}}=\prod_{s=t_0}^{t-1}\frac{\bar{n}_s^{\mathrm{policy}}}{\bar{n}_s^{\mathrm{baseline}}},\qquad O_{t+1}^j=Y_t^j.$$

All-date fertility ordering implies larger subsequent cohorts. At positive stationary limits:

$$\bar{n}^*=1/\nu,\qquad Y^*+O^*=\frac{2\bar{H}}{\bar{h}^{y*}+\bar{h}^{o*}}.$$

Larger stationary population requires less housing per adult household. This comparison requires equilibrium housing and fertility on both paths, separately from fixed-fertility welfare.

Bellman Timing

Within a four-year date, the household:

1. observes fertility taste shocks and, when eligible, chooses whether to wait or attempt a birth;
2. realizes conception success and chooses renter rooms or an owner product conditional on parity;
3. chooses consumption and next-period liquid wealth;
4. receives income and pays rent, property tax, depreciation, and any moving cost;
5. ages children independently, transitions income, and survives or leaves a warm-glow estate.

In a transition, date-t policies use the date-$(t+1)$ continuation value. The terminal value function and next-date asset price are boundary conditions; they do not replace convergence of the simulated terminal distribution.

Exact Annual Person Accounting

The annual law can be written cell by cell. For newborns,

$$N_{t+1}^{g,0} = s_{t+1}^{g,0} f_{t+1}^{g} B_{t+1} + M_{t+1}^{g,0}.$$

For interior ages $a \geq 1$,

$$N_{t+1}^{g,a} = s_{t+1}^{g,a} N_{t}^{g,a-1} + M_{t+1}^{g,a}.$$

The terminal age is top coded, so survivors from the penultimate and terminal cells both enter the last cell. Every step satisfies

$$\sum N_{t+1} = \sum N_t + B_{t+1} - \text{newborn deaths} - \text{existing deaths} + \sum M_{t+1}.$$

Exact Household-Head Accounting

Surviving heads are aged with the same survival schedule. In each cell, the target is $\rho^{g,a} N_{t+1}^{g,a}$. The accounting closure uses only one gross adjustment direction:

$$F_{t+1}^{g,a} = \max\left\{\rho^{g,a} N_{t+1}^{\mathrm{preM},g,a} - H_{t+1}^{\mathrm{surv},g,a}, 0\right\},$$

$$D_{t+1}^{g,a} = \max\left\{H_{t+1}^{\mathrm{surv},g,a} - \rho^{g,a} N_{t+1}^{\mathrm{preM},g,a}, 0\right\}, \qquad H_{t+1}^{M,g,a} = \rho^{g,a} M_{t+1}^{g,a}.$$

Hence $H_{t+1}^{g,a} = \rho^{g,a} N_{t+1}^{g,a}$ exactly. Formation and dissolution are accounting flows, not simultaneous estimated hazards.

The Within-Age Household Bridge

Let $\widetilde{G}_{t+1,a}(x)$ be the raw household transition within age a, and let $H_{t+1,a}$ be the head mass implied by persons. Then

$$G_{t+1,a}(x) = \begin{cases} H_{t+1,a}\dfrac{\widetilde{G}_{t+1,a}(x)}{\int \widetilde{G}_{t+1,a}(x')\,dx'}, & \int \widetilde{G}_{t+1,a} > 0, \\[2ex] H_{t+1,a}\Gamma_a(x), & \text{for an empty entrant age cell,} \end{cases}$$

where Γ_a is the externally specified entrant template. The operation exposes added and removed mass separately and moves no mass across ages.

A richer model would estimate economic states for newly formed, dissolved, and migrant heads rather than preserve the raw conditional composition.

Steady-State Demographic Scale

With fixed terminal survival A, newborn-survivor vector f, net migration m, and headship vector ρ, persons satisfy

$$N^* = AN^* + f\, b^*(\rho' N^*) + m,$$

where b^* is annual births per head implied by household policy.

Define

$$v_B = (I - A)^{-1} f, \qquad v_M = (I - A)^{-1} m.$$

The demographic renewal ratio is

$$\mathcal{R} = b^* \rho' v_B.$$

When fixed migration is positive, a finite steady state requires $\mathcal{R} < 1$, and

$$B^* = \frac{b^* \rho' v_M}{1 - \mathcal{R}}, \qquad N^* = v_M + B^* v_B.$$

Market and Fiscal Closure

At every date, occupied physical rooms satisfy

$$\int h(x;\pi_t)\,dG_t(x) = H_{2023}\left(\frac{P_t}{P_{2023}}\right)^{\eta}.$$

The rebated tax satisfies

$$\tau_{H,t}P_tQ_t^H = T_t\int dG_t.$$

At a stationary price,

$$r^* = (R_b - 1 + \delta_H + \tau_H)P^*.$$

Housing demand uses physical rooms. The owner-service premium affects utility but does not create additional physical supply. Conditional renter housing is also distinct from realized housing after the tenure choice.

Complete Target Fit I

Moment	Target	Model	Gap	Weight	Loss
Completed fertility, women 40–44	1.918000	1.918251	0.000251	1,425.739	0.000090
Childless share, women 40–44	0.188000	0.185867	−0.002133	17,180.744	0.078133
Mean age at first birth	26.044627	26.327334	0.282707	44.444	3.552148
First births at age 30 or later	0.260327	0.242115	−0.018213	10,000.000	3.316971
Four-year rooms contrast at first birth	0.720246	0.379786	−0.340461	137.565	15.945664
Rooms gap, 3+ versus 1–2 children	0.367700	0.359253	−0.008447	2,958.515	0.211071

Completed fertility and childlessness are CPS age-42 cohort stocks. First-birth timing uses a fixed synthetic cohort. The rooms response follows the 2019 childless risk set through matched birth and no-birth branches to 2023.

Complete Target Fit II

Moment	Target	Model	Gap	Weight	Loss
Family ownership gap	0.167662	0.176866	0.009204	14,229.591	1.205562
Ownership rate, ages 30–55	0.575472	0.546060	−0.029412	1,207.846	1.044837
Mean occupied rooms, ages 18–85	5.779970	6.710105	0.930135	11.973	10.358582
Wealth / annual gross labor earnings	6.873100	6.766352	−0.106748	6.288	0.071648
Annual bequest flow / aggregate wealth	0.008800	0.008623	−0.000177	5,165,289.256	0.161538
Old wealth-to-income p90/p50	3.448111	3.396287	−0.051824	56.960	0.152979
				Total strict loss	**36.099223**

All twelve moments enter the same active objective. The first-birth rooms row is retained as a descriptive contrast despite its non-flat prepath.

Free Parameters I

Parameter	Economic interpretation	Estimate	Search bound
β_{annual}	Patience	0.994959	$[0.9400, 0.9995]$
κ_1	First-birth logit scale	2.267631	$[0.02, 50]$
κ_C	Continuation-birth logit scale	1.765427	$[0.02, 50]$
χ	Owner-service premium	1.033945	$[0.10, 5]$
H_0	Housing-supply scale	16.381189	$[0.20, 80]$
θ_0	Bequest scale	0.563219	$[0, 8]$

No estimate is within two percent of its transformed search boundary.

Free Parameters II

Parameter	Economic interpretation	Estimate	Search bound
θ_1	Bequest wealth shift	0.099707	$[0.02, 16]$
h_m	Per-child room floor	0.315267	$[0.10, 1.80]$
h_1	First-child room jump	0.199446	$[0, 0.50]$
F_1	First-birth fixed cost	4.545167	$[0, 8]$
$\Delta\psi_{2007:2023}$	Fertility-preference change	-0.350000	$[-1.50, 0.20]$

The implied fertility intercepts are $\psi_{2007} = 0.280341$ and $\psi_{2023} = -0.069659$. The intercept change is a reduced-form residual, not a causally identified preference shock.

Assigned Household and Market Objects

Parameter	Value	Source or restriction
Model period	4 years	Timing
Entry and maximum ages	18 and 85	Timing
Risk aversion σ	2	Maintained
Financed share ϕ	0.80	20% buyer-equity restriction
Sale cost	6%	Housing literature benchmark
Owner room menu	$\{2, 4, 6, 8, 10\}$	ACS occupied-room distribution
Rental cap	6 rooms	ACS rental-size distribution
Supply elasticity η	1.75	Saiz benchmark
Entrant wealth	PSID ratio distribution	Childless renters age 18–24
Tenure taste-shock scale	0	Author restriction
Child dependence of bequests	0	Child-invariant warm glow

Historical Calibration Bridge

The 2007–2023 calibration uses five four-year dates. At each date, the observed household total and head-age marginal are imposed before the twelve maintained moments are evaluated.

- This absorbs household formation and migration that births since 2007 cannot generate by 2023.
- It makes the cross-section a dated state rather than a stationary distribution.
- Historical fertility, ownership, rooms, and price paths are holdouts unless explicitly included in the target system.
- The bridge is retired after 2023 and replaced by the person/head law.

Matching the imposed population path is an accounting identity, not evidence that the model explains U.S. household growth over 2007–2023.

Local Identification Audit

At the selected vector, the standardized residual is

$$r_j(\theta) = \sqrt{w_j}\,[m_j(\theta) - \widehat{m}_j].$$

A fresh two-sided coordinate panel constructs the weighted Jacobian $J = \partial r/\partial\theta$.

Jacobian	Rank at 10^{-3}	Condition number
All 11 free parameters	10/11	1,104.01
Ten active columns after freezing the inconsistent bequest column	10/10	391.02

The weakest right-singular vector places 91.9% of its absolute loading on θ_1. Local weak identification is concentrated in the bequest block.

Accepted Baseline Equilibrium Checks

Object	Max. absolute residual	Interpretation
Housing market	1.7943×10^{-4}	Demand relative to supply
Fiscal budget	1.1572×10^{-5}	Revenue minus equal transfers
Person identity	numerical precision	Births, deaths, and migration
Head identity	numerical precision	Deaths, formation, dissolution, migrant heads
Household/head identity	numerical precision	Household mass equals person-law head mass

The final endogenous date is also close to the independently solved terminal person distribution, head distribution, household distribution, price, rent, transfer, and fertility preference.

Slow Terminal Convergence

The dominant eigenvalue of the frozen-terminal four-year demographic block is

$$\lambda_1 = 0.960754.$$

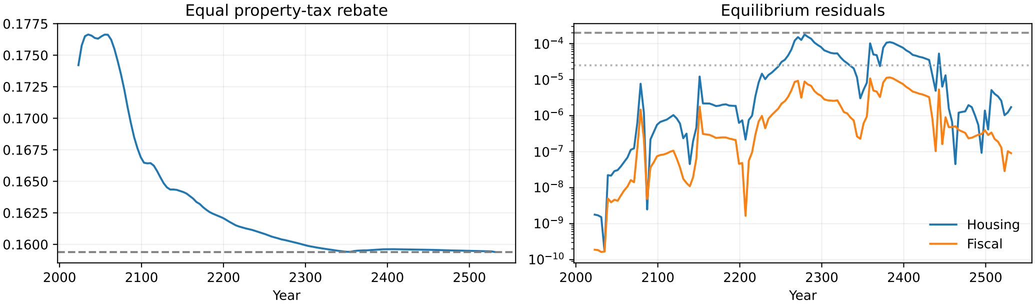
It implies

$$\text{half-life} = \frac{4\log(1/2)}{\log \lambda_1} \approx 69 \text{ years}, \qquad \lambda_1^{116} < 0.01.$$

Thus one-percent attenuation of the slow mode requires about 116 model periods, or 464 years. The accepted path uses 128 four-year periods so the simulated state approaches the terminal fixed point rather than merely inheriting it as a boundary condition.

The long horizon reflects age-structured demographic persistence, not a forecast horizon.

Baseline Transition Diagnostics



The equal rebate and equilibrium residuals are solved jointly with prices. The displayed residuals remain below their common acceptance thresholds throughout the path.

Policy Comparison Contract

A quantitative 2%–minus–1% comparison must hold fixed:

- the fitted 2023 household state and reconstructed person/head stocks;
- survival, net migration, birth sex shares, and headship;
- the housing-supply elasticity and 2023 supply anchor;
- the terminal fertility preference and all calibrated household parameters;
- the equal-rebate rule and the definitions of housing and fiscal residuals.

The comparison then separates the immediate response of prices, rents, tenure, and births from the delayed response of resident persons and household heads. No migration response or welfare ranking is part of this contract.

Interpretive Boundaries

- The first-birth rooms contrast is descriptive and has a non-flat prepath.
- The active calibration is locally weakly identified in the bequest block.
- Fixed 2023 headship is an accounting closure, not a structural formation model.
- Fixed 2100 migration in levels is essential to the positive terminal population scale.
- Static-elastic supply omits durable construction and asymmetric decline dynamics.
- One national market omits spatial sorting and local migration.
- The policy exercise reports equilibrium incidence, not welfare.

These limitations narrow the interpretation without changing the exact accounting distinction between persons, heads, and household economic states.

Selected Literature

Margin	Selected references
Lifecycle fertility	Becker and Lewis; Sommer; Doepke and Kindermann
Housing and childbearing	Lovenheim and Mumford; Dettling and Kearney; Hacamo
Housing supply and capitalization	Rosen; Roback; Saiz; Gyourko, Mayer, and Sinai
Housing tenure and user cost	Henderson and Ioannides; Sommer, Sullivan, and Verbrugge
Demography and housing demand	Mankiw and Weil; Francke et al.; Coeurdacier et al.
Spatial scale	Hsieh and Moretti; Rosen–Roback spatial equilibrium

The contribution is to combine the lifecycle housing–fertility mechanism with an explicit person-to-head demographic transition and jointly cleared housing and fiscal equilibrium.